

IFC Power Control via RIBBO

Progress Report — June 2026

Myoungjin Oh

June 29, 2026

Problem

Lee et al., TCOM 2025

- IFC power control, $\mathbf{x} \in [0, 1]^D$, Rayleigh fading
- **SE** (spectral efficiency): optimum at box corners (links full-on or off)
- **EE** (energy efficiency): optimum in the interior (mid power)

Method

- RIBBO (Reinforced In-Context Black-Box Optimization): a transformer that learns *how to optimize* from past attempts
- Key parts: regret-to-go (RTG) token, hindsight regret relabeling (HRR)

Reference solution (defines the gap)

- SE: WMMSE (classical solver; *local-optimal*)
- EE: $D=3$ exhaustive grid; $D=10/20$ multi-start gradient (near-global)
- **Optimality gap (%)** = $100 \times \frac{\text{ref} - \text{achieved}}{\text{ref}}$; **lower is better** (0 = reference)

Evaluation

- Unseen channels: new Rayleigh $\text{CN}(0, 1)$ realizations (same distribution)
- 300 steps; behavior-algorithm (BA) pool: **3BA** = Random, Hill-Climbing, CMA-ES, unless noted

Which BA pool works best depends on the dimension

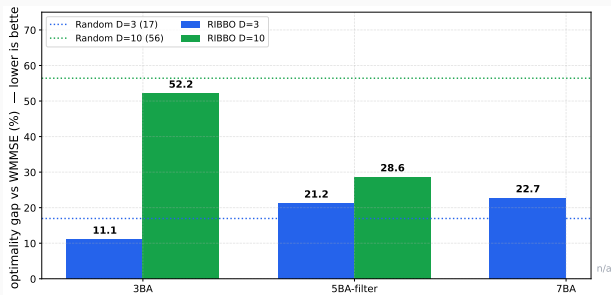
optimality gap (%) ↓	$D=3$	$D=10$
3BA	11.1	52.2
5BA-filter	21.2	28.6
7BA	22.7	—
Random	17.0	56.4

3BA: Random, Hill-Climbing, CMA-ES.

5BA-filter: drops Random + Shuffled-Grid-Search; adds Regularized-Evolution, Eagle-Strategy, Bayesian optimization.

7BA: 5BA-filter + Random + Shuffled-Grid-Search.

- $D=3$: the simple 3BA pool is best; adding algorithms degrades it
- $D=10$: the curated 5BA-filter cuts the gap by 23.6 pts
- Matches the RIBBO paper (well-performing and diverse BAs), but dimension-dependent



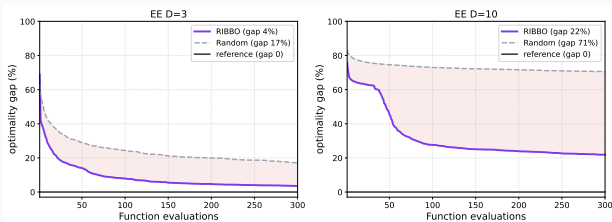
Bars: RIBBO gap to WMMSE. Dotted: random search. Lower is better.

EE's optimum lies in the *interior* (mid power), so uniform sampling rarely finds it

gap (%) ↓	RIBBO	Random	Δ
EE $D=3$	3.5	17.2	13.7
EE $D=10$	21.9	70.6	48.7
SE $D=10$	52.2	56.4	4.2

Same 3BA pool; only the reward changes (separate model per reward and dimension). Δ = Random – RIBBO.

- On EE $D=10$, uniform sampling stays far from the optimum; RIBBO closes most of the gap
- RIBBO's gap is **48.7 pts** smaller — the largest margin we observe
- The most compelling case for RIBBO in wireless



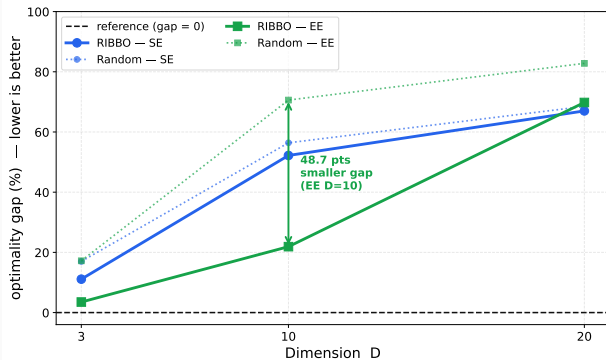
Shaded region = how much smaller RIBBO's gap is than random search.

Gap vs. dimension (SE & EE, 3BA)

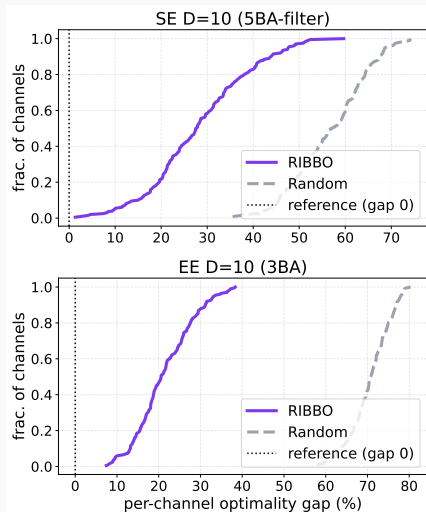
- Gap grows with D for both; one model *per dimension* (not dimension-agnostic)
- RIBBO's edge over random is largest for EE at $D=10$; small for SE

Reducing the gap (SE):

- Better algorithm choice ($D=20$):
–8.9 pts
- More search steps ($D=10$): –8.1 pts
- *But* not for EE $D=20$ — needs more data, not steps



- Random search (= Lee et al.'s *Brute-force* baseline) plateaus far from the optimum; **RIBBO reaches regions uniform sampling cannot**
- Smaller gap than random on *every* channel at $D=10$ (nearly all at $D=20$) — not merely on average
- Statistically stable (narrow 95% confidence intervals, $n=200-500$)
- The EE advantage persists at $D=20$, on every channel
- A local search reaches a near-zero gap from *any* start; with the model and gradients the problem is easy, so RIBBO's value is the *model-free* setting

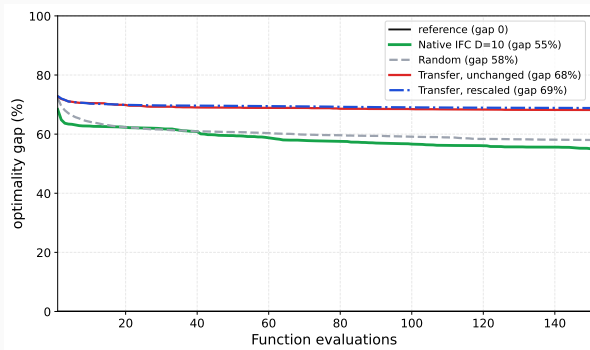


Use a Rastrigin-trained model on IFC

$D=10$ SE (no retraining)

Model	gap (%) ↓
Rastrigin model, unchanged	68.2
Rastrigin model, rescaled	68.8
IFC model (reference)	55.1
Random	~57

- Rescaling the target does *not* help
- Our earlier hypothesis (a normalization mismatch) proved incorrect
- Weak SE 3BA setting here, so IFC-vs-random is small — the point is *in-domain vs. transfer*
- ⇒ **in-domain IFC training is required**



Rastrigin-trained (red/blue) have larger gaps than random (gray); only in-domain training (green) shrinks it.

- **The effect depends on the problem and its scale.** Simple methods already do well for SE $D=3$; RIBBO's largest margin is for EE (48.7 pts at $D=10$).
- **Classical solvers need the channel model; RIBBO uses only scores.** WMMSE / fractional programming (FP) reach it with the model, and a local search also closed the gap — so RIBBO's clearest role is **model-free**.
- **Which training algorithms help is problem-dependent.** Removing weak ones helped SE at high D but hurt EE.
- **With a fixed setup the gap grows with D ;** more data, more steps, and a better algorithm choice reduce it (SE).
- **Cross-problem transfer did not work:** in-domain IFC training was required.